Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Field	Details
Testing Tool Used	Flask Local Server, Web Browser
Type of Testing	Response Time Testing, Validation Testing, Functional Testing
Target Module / API	Crop Recommendation Endpoint (/predict)
Test Environment	Windows 10, Python 3.x, Flask Development Server
Test Date	07 July 2026

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits crop prediction request.	1	5	Crop recommendation displayed successfully.
2	Multiple users submit prediction requests one after another.	10	30	System remains responsive.
3	Concurrent crop prediction requests.	25	60	Stable performance without failures.
4	Invalid or incomplete input values submitted.	5	20	Validation messages displayed correctly.

Metric	Target Value	Actual Value	Status	Remarks
Average Response Time	< 2 sec	1.3 sec	Pass	Fast prediction response.
Maximum Response Time	< 5 sec	2.7 sec	Pass	Stable during testing.
Throughput	10 req/sec	12 req/sec	Pass	Meets expected performance.
Error Rate	< 1%	0%	Pass	No prediction failures .
CPU Utilization	< 80%	45%	Pass	Efficient CPU usage.
Memory Utilization	< 80%	52%	Pass	Stable memory usage.